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Abstract: We document that between 50% and 70% of changes in the U.S. wage structure over the last four decades are accounted for by relative wage declines of worker groups specialized in routine tasks in industries experiencing rapid automation. We develop a conceptual framework where tasks across industries are allocated to different types of labor and capital. Automation technologies expand the set of tasks performed by capital, displacing certain worker groups from jobs for which they have comparative advantage. This framework yields a simple equation linking wage changes of a demographic group to the task displacement it experiences. We report robust evidence in favor of this relationship and show that regression models incorporating task displacement explain much of the changes in education wage differentials between 1980 and 2016. The negative relationship between wage changes and task displacement is unaffected when we control for changes in market power, deunionization, and other forms of capital deepening and technology unrelated to automation. We also propose a methodology for evaluating the full general equilibrium effects of automation, which incorporate induced changes in industry composition and ripple effects due to task reallocation across different groups. Our quantitative evaluation explains how major changes in wage inequality can go hand-in-hand with modest productivity gains.

Language: English**Subjects:** ;**plink:** <https://research.ebsco.com/linkprocessor/plink?id=939bd2f1-018a-33f7-a90a-6507ca507ced>